Self-Conjugation as a Structural Research Program

Master Meta-Draft: Scope, Formal Objects, an 8D Alternative-Slice Compiler, and Structural Topological Meta-Fields

Working research document

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Document status. This is a scope-and-architecture master meta-draft, not a completed mathematical paper. It distinguishes established algebraic facts, candidate definitions, engineering contracts, and speculative cross-domain hypotheses. No physical interpretation or platform integration is declared verified merely because it is written here.

Abstract

This document proposes a disciplined research program for the Self-Conjugation (SC) framework: a typed transformation process that compares a structure with a declared conjugate, dual, contrast, or transformed representation; records the resulting invariant and residue separately; and uses retained information and external constraints to determine what can be reconstructed or changed. The program is paired with an 8-dimensional (8D) alternative-slice compiler motivated by the octonions within the Cayley–Dickson construction, and with a common representation for structural topological meta-fields that can be translated into Perchance generators and other generator hubs.

The central engineering opportunity is not a claim that all higher-dimensional structures reduce to octonions. It is a compiler and verification discipline that makes domains, allowed transformations, preserved invariants, residual differences, provenance, and acceptance tests explicit. A partial-category interpretation is viable only if its objects, morphisms, composition domain, identities, and associativity conditions are specified and checked. Likewise, a non-tautological symmetry measure must be built from well-defined algebraic or geometric data and a declared sampling measure; a ratio that can exceed one is not a Shannon entropy. Algebraic structural entropy is kept separate from thermodynamic entropy unless an explicit physical model supplies a bridge.

The paper lays out a phased implementation plan: formal contract; exact arithmetic and reference tests; alternative-slice representation and composition checks; Self-Conjugation operator semantics; platform-neutral meta-field intermediate representation; Perchance proof-of-concept; portable adapters; cold-start and failure-injection verification; and only then deeper entropy and physics research. Acceptance gates, risks, document structure, and project-prioritization rules are included so the work can advance without displacing separately committed benchmark or case-work tracks.

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1 Purpose, scope, and intended contribution

1.1 Research question

Can a transformation-and-verification architecture expose what remains invariant, what has been lost, and what new structures become available when a system compares and transforms its own representation, while making the result portable across algebraic compilers and AI generator hubs?

1.2 Proposed contribution

The proposed contribution has four linked parts:

1. a typed Self-Conjugation pipeline separating transformation, invariant extraction, residue, information retention, and realization;
2. an 8D alternative-slice compiler that tests the actual algebraic closure and composition conditions instead of relying on coordinate-count heuristics;
3. a structural meta-field intermediate representation (IR) that can preserve graph/topological relationships, constraints, provenance, and evidence across platform adapters;
4. an independent verification protocol for generator composition, including reproducible cold starts, perturbation tests, artifact hashing, and bounded verdicts.

1.3 Scope boundary

The project is both a mathematical investigation and a software-architecture program. The former defines and proves claims; the latter implements declared contracts. A successful implementation is evidence about that implementation only. It does not automatically prove an ontological interpretation, establish a physical mechanism, or validate the entire theoretical program.

1.4 Non-goals for the first release

The first release does not claim to derive new physics, establish a universal law of structural entropy, prove that every high-dimensional algebra factors through an 8D slice, or construct a fully general category of all possible generator workflows. It does not require a private or undocumented platform API. It does not alter the separately tracked Omega benchmark Phase 1 or claim that publication rank validates the internal mathematical architecture.

2 Status vocabulary and epistemic contract

Every major statement and artifact should carry one of these states:

Status	Meaning
Established fact	Supported by a standard definition, proof, or reliable source within its stated scope.
Derived	Follows from explicitly declared definitions and assumptions; proof or derivation is retained.

Proposed	A design or mathematical definition offered for testing; not yet established as useful or unique.
Analogy	A structural resemblance across domains without a proven shared mechanism.
Implemented	Present in code or a data artifact; does not alone establish correctness.
Tested	A named test passed on a specified input, version, and environment.
Verified	A claim passed a specified acceptance contract with retained evidence and a challenge path independent of the claim-generating mechanism.
UNVERIFIED	Evidence, source material, a definition, or a required test is missing.

2.1 Non-self-certification rule

A transformation may generate a claim; an invariant may support it; neither the transformation nor the invariant supplies the full warrant by itself. The system that proposes a correction cannot be the sole authority that certifies the correction. A missing required artifact or acceptance test yields UNVERIFIED, not a success by default.

2.2 Relation to Gödelian Epistemic Θ-Integrity

The proposed governance layer preserves distinctions among claim, warrant, evidence, verdict, and unresolved residue. The Gödelian reference is a structural analogy concerning limits on self-justification; this project does not claim to be a direct application or consequence of Gödel's incompleteness theorems. Changes to the evaluator or acceptance rules must be versioned, and previous run records must not be retroactively rewritten.

3 Architecture at a glance

3.1 Three interacting layers

1. **Transformation layer:** Self-Conjugation, typed subtraction, alternative-slice operations, and candidate invariant extraction.
2. **Representation layer:** structural meta-fields, typed relations, topological data, evidence, provenance, and versioned intermediate representations.
3. **Governance layer:** acceptance contracts, cold-start replay, failure injection, bounded verdicts, and retained artifacts.

3.2 System flow

Input structure and context → declared transform → invariant / residue split → typed meta-field IR → compiler or platform adapter → independent acceptance tests → bounded verdict and retained provenance

3.3 Relationship to MRMD

The Meta-Realized Meta-Cairn DropBox (MRMD) should initially be the persistence format for raw anchors, cairns, meta-cairns, candidate relations, evidence, and open obligations—not a parallel UI project. The original expression remains intact; interpretation and links can evolve without changing the historical record. A candidate relation between two cairns is not treated as identity or equivalence by default.

4 Self-Conjugation: typed transformation, invariant, and residue

4.1 Why the name requires a type contract

The complex-conjugation example is informative but not a general definition. For $z = a+bi$, complex conjugation gives $\bar{z} = a - bi$ and $z\bar{z} = a^2 + b^2 = |z|^2$. The norm is invariant under phase changes, so the scalar $|z|^2$ alone does not preserve phase. A magnitude plus an independently retained phase can reconstruct z (subject to the representation's conventions); the magnitude alone cannot.

The general framework therefore uses a domain-specific conjugation or contrast map, not a universal claim that every self-conjugation is an involution or a scalar norm operation.

4.2 Candidate typed definition

Definition 4.1 (Typed Self-Conjugation record)**.** *Let $\mathcal{F}$ be a declared space of structures and let $\mathcal{D}$ specify the domain, semantics, and admissible transformations. A Self-Conjugation instance is a tuple*

$$\mathsf{SC}_{\mathcal{D}} = (J_{\mathcal{D}}, \delta_{\mathcal{D}}, I_{\mathcal{D}}, \mathrm{Res}_{\mathcal{D}}, K_{\mathcal{D}}, W_{\mathcal{D}}),$$

where $J_{\mathcal{D}}$ maps a structure to a declared transformed representation; $\delta_{\mathcal{D}}$ compares the original and transformed representations; $I_{\mathcal{D}}$ extracts a candidate invariant; $\mathrm{Res}_{\mathcal{D}}$ *records a typed difference or residue; $K_{\mathcal{D}}$ defines any reconstruction/realization operation; and $W_{\mathcal{D}}$ is the external constraint and evidence context.*

A run may be written as

$$C_n = J_{\mathcal{D}}(F_n), \tag{1}$$
$$I_n = I_{\mathcal{D}}(F_n, C_n), \tag{2}$$
$$r_n = \mathrm{Res}_{\mathcal{D}}(F_n, C_n), \tag{3}$$
$$F_{n+1} = U_{\mathcal{D}}(F_n, I_n, r_n, \Delta_n^{\perp}, W_n). \tag{4}$$

Here $\Delta_n^{\perp}$ denotes retained transverse information where the domain supports such a notion. It must not be added decoratively: the schema must state what it measures, how it is obtained, and which reconstruction ambiguity it resolves.

4.3 Properties to test, not assume

- **Involution:** $J_{\mathcal{D}}^2 = \mathrm{id}$ only when proved for the declared domain.
- **Invariance:** a candidate I must specify exactly which transformation leaves it unchanged.

- **Residue typing:** a formal residual, an evidentiary gap, and a physical entropy term are different data types.
- **Reconstruction:** state whether K is exact, approximate, unique, partial, or impossible under stated information loss.
- **Admissibility:** a rule update is accepted only against versioned external constraints and an independently executable test.

4.4 Generative subtraction and self-conjugation

The proposed sequence is

$$U \xrightarrow{\ominus Y} H_U(Y) \xrightarrow{J_{\mathcal{D}}} I_Y \xrightarrow{K(I_Y,\Delta^{\perp},\rho,W)} F'.$$

The notation $U \ominus Y = H_U(Y)$ is a proposed, context-relative operation: the removed or contrasted element may expose a boundary or a distinction. This does not imply that subtraction always creates a useful invariant or new capability. The relevant questions are empirical within a model and algebraic within a formal system: what was removed, what new distinction appeared, what information survived, and what capability can now be demonstrated?

4.5 Research tests

Acceptance criterion 4.1 (SC round trip and loss accounting)**.** *For each exemplar domain, publish an exact input, transform, invariant, residue, retained side-channel, reconstruction attempt, and error metric. The test must demonstrate at least one case where the invariant is preserved while another feature is lost, and at least one case where the retained side-channel is necessary to recover the target.*

5 Cayley–Dickson tower and the proposed 8D alternative-slice compiler

5.1 Mathematical baseline

Let $\mathcal{A}_n$ denote the n-th real Cayley–Dickson algebra, with real dimension $D_n = 2^n$ for the usual doubling sequence. The initial sequence is $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$, and the 16-dimensional sedenions. The quaternions are noncommutative; the octonions are nonassociative but alternative and form a normed division algebra; the standard sedenions are nonassociative, nonalternative, power-associative, and contain zero divisors [1,2]. These are algebraic facts, not evidence of a physical phase transition.

5.2 What an 8D slice means in the compiler

The proposed slice object is an explicitly represented 8D subalgebra $S \subseteq \mathcal{A}_n$ whose basis, multiplication table, embedding, conjugation, and relevant identities are specified. Its validity must be computed from closure and operation tables. Merely having at most eight nonzero coordinates does not prove that a vector collection is closed under multiplication or lies in a valid octonion subalgebra.

5.3 Objects, morphisms, and partial composition

A candidate partial category Slice has objects containing a declared slice, source/target schema, invariants, constraints, and version. Morphisms are explicitly typed transformations that preserve the declared structure to the required level. Composition is allowed only when the codomain/source contracts match and the composed transformation satisfies the slice and output constraints.

Definition 5.1 (Candidate slice morphism)**.** *A morphism $f : X \to Y$ is a recorded map together with a contract specifying its domain, codomain, preserved properties, permitted loss, residue type, and acceptance tests. The compiler admits $g \circ f$ only when interface compatibility and the declared structural predicates pass.*

A partial-category claim requires identity morphisms and associativity whenever the relevant composites are defined. Function composition of valid maps is associative; this does not make octonion multiplication associative. If the intended operation is multiplication itself rather than composition of structure-preserving maps, the associator must be represented explicitly and the structure may need a weaker categorical or higher-categorical treatment. The implementation should not label an arbitrary typed workflow graph a category until its axioms pass.

5.4 Compiler contract

Check	Required evidence
Domain and basis	Explicit algebra identifier, dimension, basis ordering, structure constants, and embedding map.
Slice closure	For basis elements e_i, e_j in the proposed slice, multiplication remains in the slice; test against exact or controlled symbolic coefficients.
Identity and conjugation	Declared unit/conjugation semantics and applicable identity tests.
Property scope	Distinguish alternative, associative, power-associative, normed, and division properties; do not bundle them into a vague "stable" label.
Composition	Source/target compatibility, closure after composition, and associativity for admitted composites.
Loss and residue	Any projection or truncation logs which coordinates/relations were dropped and what could not be reconstructed.
Evidence	Retained test vectors, expected outputs, implementation version, and reproducible failure examples.

5.5 Initial algebraic test matrix

The test engine should verify known properties with a canonical multiplication implementation before testing compiler policy. It should include basis-level and seeded random tests for multiplication, conjugation, norm identities where applicable, associators, alternative identities, zero-divisor pairs, subalgebra closure, and slice-to-slice maps. Numerical floating-point equality is not sufficient for exact algebraic claims when rational/integer structure constants can be used.

6 Structural topological meta-fields

6.1 Platform-neutral intermediate representation

The meta-field IR is a versioned serialization format that describes a structural object and how it may be transformed or composed. A minimum record should contain:

- identity: schema version, object ID, source, timestamp, implementation version, content hash;
- domain: algebra/model family, basis or feature vocabulary, units and type declarations where relevant;
- topology: nodes, edges, higher-order relations, boundary data, connectivity, and any explicitly computed topological descriptors;
- operators: named transformations with typed inputs/outputs and preconditions;
- invariants and residues: what is compared, preserved, lost, generated, or unresolved;
- admissibility: allowed slices, objective, forbidden transitions, acceptance tests, and failure modes;
- provenance: source artifacts, dependency versions, seed, and run trace;
- epistemic status: observed, analogy, proposed, derived, implemented, tested, or verified.

6.2 Topological claims need a concrete representation

A graph, simplicial complex, cell complex, or other topological object must be selected for each use case. Terms such as "curvature", "field", "fiber", "torsion", and "homology" should be used computationally only after the underlying spaces, maps, coefficient structures, and operators are defined. A visual graph of concepts is useful, but it does not by itself instantiate a differentiable manifold, a fiber bundle, or a physical field.

6.3 Why the IR is the leverage point

The same canonical object can be compiled into different user interfaces without collapsing the underlying structure into platform-specific prompts. Platform adapters become renderers and runtime bindings around a common core. This is the route to reusing the same typed meta-fields in Perchance, local notebooks, other generator hubs, documentation systems, and future agent workflows.

7 Non-tautological symmetry descriptors and structural entropy

7.1 Correcting the preliminary index

A preliminary expression used $r = \dim \mathrm{Aut}(A)/\dim(A)$ and $-r \log_2 r$. This is not a Shannon entropy in general: for octonions, $\dim G_2 = 14$ while the real vector-space dimension is 8, so $r > 1$ and the expression becomes negative. The denominator is not a probability normalizer. The preliminary table also assigned an automorphism-group dimension of 3 to the standard sedenions without a sufficient basis; standard treatments report $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$ for the standard construction, which has Lie-group dimension 14 because S_3 is finite [3]. Exact conventions should be rechecked before any dataset is frozen.

7.2 Descriptor before entropy

A safer first metric is a normalized symmetry-dimension descriptor. For a real D-dimensional algebra whose automorphisms act orthogonally, define the ambient continuous rotation budget $d_{\mathrm{amb}} = D(D-1)/2$ and

$$s(A) = \frac{\dim \mathrm{Lie}(\mathrm{Aut}(A))}{D(D-1)/2}, \qquad D \geq 2.$$

This is a candidate symmetry-density descriptor, not entropy. Its assumptions, symmetry group, field, and representation must be recorded. It does not encode the zero-divisor locus, orbit sizes, stabilizers, associator structure, or information loss on its own.

7.3 Zero-divisor geometry

For algebra A, begin with the explicitly typed relation

$$Z_A = \{(x, y) \in A \times A : xy = 0,\ x \neq 0,\ y \neq 0\}.$$

To report an algebraic-variety dimension, specify whether the zero locus is projectivized, which scalar field is used, whether norm constraints are imposed, and whether ordered pairs or equivalence classes are counted. A finite list of representative zero-divisor pairs is not the same object as the dimension of a zero-divisor variety. The number "9" appearing in preliminary notes is therefore treated as UNVERIFIED until its exact definition and derivation are documented.

7.4 When an entropy becomes legitimate

An entropy requires a defined distribution or measure. One candidate is the entropy of a predeclared structural profile under a specified sampling measure μ:

$$H_\mu(X) = -\sum_{b \in \mathcal{B}} p_\mu(b) \log p_\mu(b),$$

where $\mathcal{B}$ is a fixed set of bins/strata and p_μ is the distribution induced by an explicit sampling procedure. Another candidate uses orbit classes or admissible transformation outcomes. These are research candidates: bins, measure, normalization, invariance requirements, and sensitivity to representation must be pre-registered and tested. A count of properties that have already been declared "lost" is tautological if the same labels define both the input and the outcome.

7.5 Minimum validation suite

1. compare candidate metrics against known algebras and exact automorphism data;
2. test robustness under basis change and equivalent representations;
3. measure sensitivity to sampling choices and binning;
4. use held-out transformations to test whether the index predicts a nontrivial structural event;
5. report null models and counterexamples, including cases where symmetry dimension and zero-divisor structure do not move together.

8 Boundary between algebraic structure and physical thermodynamics

The notation "structural entropy" may describe information, complexity, or a distribution over algebraic profiles, but it is not automatically thermodynamic entropy. A formula such as $\Delta S_{\text{thermal}} = -\alpha \Delta S_{\text{struct}}$ or $\Delta Q = T \Delta S_{\text{thermal}}$ requires a physical state space, a statistical ensemble, units, a coupling model, and conditions under which the thermodynamic identity applies. Those elements are not supplied by the Cayley–Dickson construction alone.

The Second Law remains a stipulated constraint in Case Ξ–001, not a theorem derived from a declining algebraic dimension. The proposed null substrate, $\Delta_{1/2}$, "The Hum", and any relation to cosmological or quantum phenomena remain hypotheses until the objects, physical mechanism, observables, competing models, and discriminating predictions are defined. Voyager plasma waves and pulsar-timing evidence may be separate physical case studies; shared vocabulary does not establish a common cause.

9 Applications and R&D value

9.1 High-value near-term applications

Application	Value proposition	Evidence required
Algebraic reference compiler	Makes Cayley–Dickson operations, properties, and loss conditions inspectable.	Exact reference arithmetic, canonical examples, test coverage, independent expected results.
Slice-valid workflow composition	Prevents an agent/generator chain from being admitted on a superficial dimensional heuristic.	Closure checks, typed interfaces, associativity/identity checks for the claimed category.
Structural meta-field IR	Preserves reusable structure across UI and generator hubs.	Schema validation, lossless round trips where promised, explicit lossy conversions.
Correction and provenance auditing	Makes each proposed change inspectable and challengeable.	Immutable run records, failure injection, cold-start replay, independent verdict criteria.
MRMD research memory	Turns realizations into linked, recoverable research objects rather than a pile of notes.	Byte-preserved source anchors, typed links, provenance reconstruction, contradictory interpretation support.
Physics/quantum exploration	Creates a disciplined staging area for mathematical structures that might inform a physical model.	Formal map to physical variables, units, empirical discriminators, peer-reviewed independent review.

9.2 What is valuable even if the physical hypothesis fails

The compiler, typed IR, transformation ledger, and acceptance machinery retain engineering value without a cosmological or quantum interpretation. They can make generator composition more reliable, support explainable transformations, and expose where information has been discarded. This separation protects the practical work from dependence on unresolved metaphysical claims.

10 Perchance and other generator-hub integration

10.1 Integration strategy

Use a platform-neutral core and add adapters only after the intermediate representation and acceptance tests stabilize. For Perchance, the initial route should use documented generator composition/import, normal share URLs, downloads where appropriate, and embeds. Perchance documentation describes importing reusable generators, sharing URLs, downloading generators, and embedding a generator with an iframe. A 2023 platform guide states that Perchance did not then provide an official general API; that status must be rechecked before relying on any API in 2026, and a private or inferred endpoint must not be assumed stable [7, 8]. The platform also warns that renaming a generator can break importers, so release URLs should be treated as dependencies and stable versions should be copied/forked rather than renamed in place. If a custom API is required, isolate it as a separately deployed adapter and treat it as a distinct integration project.

10.2 Reference architecture

1. **Core model:** canonical JSON-like IR, schema validator, deterministic compiler functions, provenance writer.
2. **Algebra package:** exact structure constants, slice objects, composition checker, invariant/residue accounting.
3. **Generator adapter:** input/output translation, prompt and list rendering, platform import/embed settings, graceful unsupported-feature behavior.
4. **Auditor:** cold-start execution, seeded repetition, schema conformance, artifact hashing, and perturbation tests.
5. **Publication edge:** user-controlled creation/publishing of generator pages and share links. The local compiler can prepare artifacts but cannot claim to have published or externally validated them unless that action actually occurred.

10.3 Adapter contract

Every adapter declares: supported IR version; supported object and relation types; unsupported or lossy features; input/output schema; seed behavior; persistence and privacy assumptions; link/embed behavior; acceptance suite; dependency versions; and known failure modes. Platform-specific functionality must not leak into the canonical math layer.

10.4 A minimal Perchance proof-of-concept

The first Perchance demonstration should be deliberately small: one generator imports or embeds a compiled meta-field; one operation chain is generated from a fixed seed/input; the generator prints the source object ID and compiler version; an independent checker verifies the output against the IR and records a content hash. The test is integration proof only; it does not validate the broader Self-Conjugation theory.

11 Verification and independent challenge

11.1 Cold-start protocol

A cold-start run begins from a fresh execution context with no hidden in-memory state. It loads only declared inputs, validates schema and hashes, replays the transformation, and checks output against the versioned contract. A replay is not considered independent if it silently reads previous output as input or reuses mutable state that is absent from the recorded manifest.

11.2 Failure-injection suite

At minimum, inject: removed evidence; mutated invariant; changed basis order; invalid slice embedding; incompatible morphism target; missing imported generator; unknown IR version; nondeterministic seed handling; truncated provenance; and a corrected output that violates the stated objective. Each mutation must either be detected or be explicitly listed as an untested boundary. A passing normal-path run is not enough.

11.3 Bounded verdicts

- **PASS:** required artifacts exist and the scoped acceptance test passes.
- **FAIL:** the test contradicts the claim or detects a prohibited transformation.
- **UNVERIFIED:** required evidence, definitions, or tests are missing.
- **PARTIAL:** only when a predefined multi-part contract explicitly permits partial satisfaction; specify the exact passed and failed components.

A local test pass must not be promoted into a global guarantee. Version changes may update the present status but must preserve earlier run records.

12 Phased development plan and acceptance gates

The sequence is dependency-driven. Phase numbers are gates rather than promises of calendar duration; each phase opens only when its stated exit conditions pass.

12.1 Phase 0 – Freeze the contract and repair unsupported claims

Deliverables: canonical vocabulary; typed distinction between algebraic residue and epistemic gap; source ledger; correction of the automorphism-group table; retirement of the invalid $-r \log r$

index as an entropy measure; definition of which sedenion zero-divisor object is being measured.
Exit gate: no key metric or algebra property is both undefined and treated as a result.

12.2 Phase 1 – Exact Cayley–Dickson reference engine

Deliverables: reference multiplication for $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}, \mathbb{S}$; basis/unit tests; conjugation and norm tests; associator and alternative-identity tests; known zero-divisor examples; deterministic test vectors.
Exit gate: expected outputs are sourced or derived independently, and deliberate mutation of a structure constant causes a regression failure.

12.3 Phase 2 – 8D slice registry and composition compiler

Deliverables: slice schema; embedding maps; closure verifier; typed morphism contract; partial-composition decision procedure; residual/loss record.
Exit gate: all admitted compositions pass the declared predicates; rejected compositions include a reproducible reason; category axioms are tested before the system is called a category.

12.4 Phase 3 – Self-Conjugation engine

Deliverables: domain-typed J, comparison, invariant, residue, side-channel and realization interfaces; one exact worked example and one lossy example; versioned external constraint input.
Exit gate: the system correctly distinguishes invariant retention from full-state reconstruction and does not treat a candidate invariant as proof of identity.

12.5 Phase 4 – Structural meta-field IR and MRMD binding

Deliverables: schema and validation; records for a raw anchor, cairn, meta-cairn, relation, evidence item, and test run; byte-preserved anchors; provenance graph.
Exit gate: two cairns can be linked as a candidate correspondence without being merged, contradictory interpretations survive, and a derived record's provenance can be reconstructed.

12.6 Phase 5 – Perchance integration proof

Deliverables: one imported or embedded generator, shareable output, adapter manifest, seed behavior, source/version reference, and independent output check.
Exit gate: cold-start replay and failure injection pass on the actual published generator; no assumption of API access is needed for the minimum proof.

12.7 Phase 6 – Cross-hub adapters and external use

Deliverables: generic adapter interface; at least one second hub selected on explicit feasibility criteria; compatibility and unsupported-feature matrix; portability report.
Exit gate: core IR and algebra rules remain platform-neutral, with any conversion loss logged. External user response tests product usefulness, not mathematical truth.

12.8 Phase 7 – Non-tautological symmetry and entropy research

Deliverables: validated automorphism data; explicit zero-divisor variety conventions; candidate normalized descriptors; preregistered sampling/bins; baseline and null models.
Exit gate: the measure is representation-robust within stated limits and predicts a held-out structural outcome better than a declared baseline. Until then the term "entropy" remains provisional.

12.9 Phase 8 – Physical bridge, only if justified

Deliverables: physical model, state variables, units, ensemble, coupling assumptions, conservation accounting, competing explanations, and observable discriminator.
Exit gate: an independent domain expert can evaluate the derivation without relying on the symbolic vocabulary alone. No physical or thermodynamic claim is promoted from analogy.

13 Priority matrix and portfolio management

Priority	Workstream	Why now	Stop/exit condition
P0	Correct baseline, notation, source/status ledger	Prevents a false theorem or invalid metric from becoming infrastructure.	Every central claim has a type, source/derivation, and open obligation.
P1	Exact reference algebra + tests	Required by every downstream slice and entropy claim.	Canonical examples and mutation tests pass.
P1	Alternative-slice compiler MVP	Reusable technical core; tests the most concrete computational claim.	Closure and partial-composition contracts pass.
P2	Self-Conjugation typed API	Makes invariant/loss accounting reusable beyond the algebra project.	Round-trip and loss examples pass.
P2	Meta-field IR + MRMD records	Creates the reusable persistence layer for case work and generator composition.	Schema/provenance acceptance suite passes.
P3	One Perchance proof-of-concept	Tests actual integration without committing to platform-specific expansion.	Cold-start replay and failure injection pass.
P3	One second-hub adapter	Establishes portability after the core stabilizes.	Explicit feature parity and loss report completed.
P4	Structural symmetry descriptor/entropy	Potential research payoff, but depends on corrected invariants and exact code.	Predeclared measure survives held-out testing.
P5	Thermodynamic / quantum / cosmological bridge	High intellectual reach, but evidence and modeling prerequisites are largest.	Formal and empirical bridge exists; otherwise retain as hypothesis.

13.1 Protected boundaries with other projects

- Keep the separately committed Omega benchmark Phase 1 unchanged unless the project owner explicitly changes that commitment.

- Treat Ladder-V publication as a user-side ship boundary. Publishing one composition tests discoverability and use; rank is not a validator for the internal mathematical framework.
- Let MRMD support existing corpus and case work. Do not start a separate interface project before the source records, schema, and acceptance tests are sufficient.
- Do not reconstruct missing Word/Excel master files from fragments and then label them the originals. If source files are absent from the runtime, report the boundary and request or mount the actual files before modifying them.

14 Document structure and intended publication package

14.1 Master-paper table of contents

The present file is the governing master meta-draft. A publication-ready version can be split into the following companion artifacts:

1. **Paper A – Formal Self-Conjugation:** definitions, invariants, residues, reconstruction conditions, worked proofs, counterexamples.
2. **Paper B – Alternative-Slice Compiler:** Cayley–Dickson reference algebra, slice embeddings, partial composition, category axioms, implementation and verification results.
3. **Paper C – Structural Symmetry Descriptors:** automorphism data, zero-divisor geometry, sampling measure, candidate entropy, sensitivity analysis and null models.
4. **Engineering note – Meta-field IR and hub adapters:** schema, adapter contract, Perchance proof, cold-start replay, limitations, portability matrix.
5. **Case file – Physical bridge obligations:** retain Ξ–001 claims, sources, unknowns, and the Second Law constraint without turning analogies into findings.

14.2 Research artifacts expected from each phase

Each completed phase should produce an immutable run manifest, source and dependency versions, the exact acceptance contract, retained test inputs/outputs, failure-injection results, limitations, and a short decision record. Figures and tables should be regenerated from the versioned data rather than manually transcribed. The Word docket and Excel register should be updated only from their actual authoritative files; a reconstruction made from excerpts is a new derivative artifact, not a replacement original.

15 Open questions and falsifiable obligations

1. What precise object does $J_{\mathcal{D}}$ act on in each domain: state, rule, representation, or transformation history?
2. When does $J^2 = \mathrm{id}$ hold, and what replaces involution when it does not?
3. Which information is compressed into an invariant, and which residue or side-channel is minimally sufficient for reconstruction?
4. Which admissible 8D slices occur in each target algebra, how are embeddings found, and what closure theorem or complete finite test supports them?
5. Is the intended structure a genuine partial category, a typed graph, a restriction system, or a weaker structure with associator/coherence data?

6. Which symmetry invariants remain robust under basis change and equivalent presentations?
7. What sampling measure gives the proposed structural entropy a non-tautological interpretation?
8. What exact physical model would connect an algebraic descriptor to thermodynamic entropy, quantum observables, or cosmological claims?
9. Which platform features are truly shared across generator hubs, and which require explicit lossy adaptation?

16 Conclusion

The highest near-term value is a falsifiable compiler-and-verification stack: exact algebraic operations; explicit alternative-slice predicates; typed Self-Conjugation records; a platform-neutral meta-field IR; and independent acceptance tests. This stack remains useful even if speculative physical interpretations do not survive. Symmetry entropy and physics bridges are valuable research directions, but they depend on precise definitions, reference computations, declared measures, and discriminating tests. The disciplined sequence is therefore: formalize, implement, perturb, replay, integrate, and only then generalize.

References

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[2] G. Moreno, "Sedenions: algebra and analysis," *Applied Mathematics and Computation*, vol. 115, nos. 2–3, pp. 77–88, 2000. DOI: 10.1016/S0096-3003(99)00140-X.

[3] "An exceptional G(2) extension of the Standard Model from the correspondence between Cayley–Dickson algebras and automorphism groups," *Scientific Reports*, 2021. DOI: 10.1038/s41598-021-01814-1. Used here for a reported automorphism-group relation, not as evidence that the proposed physical interpretation is established.

[4] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.

[5] C. E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal*, vol. 27, pp. 379–423 and 623–656, 1948.

[6] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.

[7] Perchance, "Embed Generator" and generator-sharing documentation, available at `https://perchance.org/embed-generator`. Accessed 11 October 2026. Platform behavior should be rechecked before each integration release.

[8] Perchance, "Create a Random Generator" (platform introduction / sharing and importing overview), historical mirrored documentation available at `https://archive.is/P0ogl`. Accessed 11 October 2026. Treated as descriptive documentation, not a guarantee of an official API.

A Machine-readable run manifest: minimum fields

A run manifest should contain at least:

```
run_id
schema_version
compiler_version
algebra_id
basis_order
slice_id and embedding
input_artifact_hash
transform_contract
invariant_definition and value
residue_type and value
retained_side_channels
platform_adapter and version
seed / runtime / dependencies
acceptance_contract
failure_injection_results
verdict: PASS | FAIL | UNVERIFIED | PARTIAL
known_limits
prior_run_reference
```

B Suggested repository layout

```
self-conjugation-rd/
  paper/
    self_conjugation_master_meta_draft.tex
    references.bib                  # optional later extraction
  spec/
    self_conjugation.schema.json
    alternative_slice.schema.json
    meta_field_ir.schema.json
  core/
    cayley_dickson_reference/
    slice_registry/
    self_conjugation/
    meta_field_ir/
  adapters/
    perchance/
    hub_template/
  tests/
    algebra_reference/
    slice_closure/
    category_axioms/
    failure_injection/
    cold_start_replay/
```

```
evidence/
  run_manifests/
  source_ledger/
mrmd/
  inbox/
  cairns/
  meta_cairns/
  relations/
```